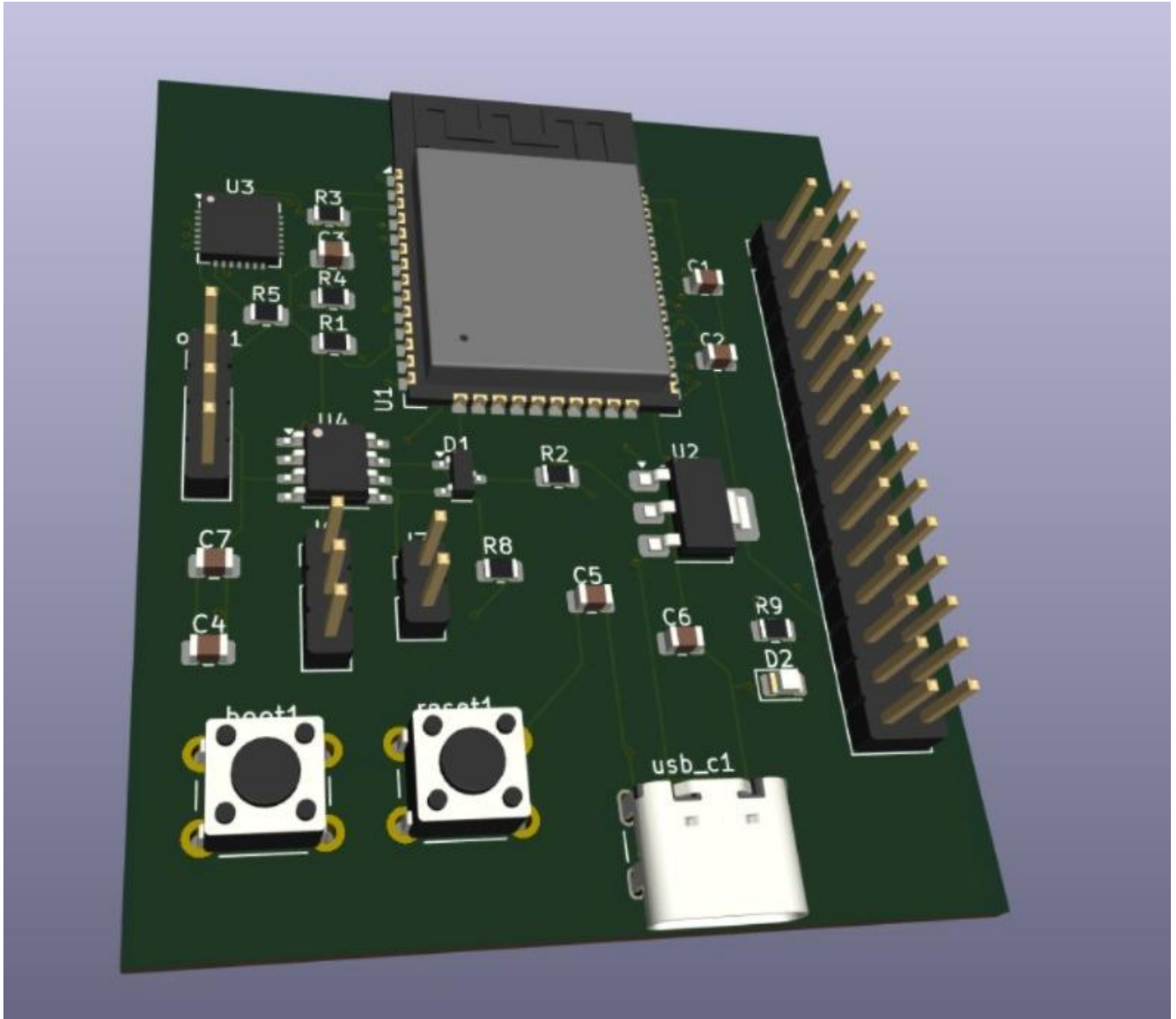


ESP32 DEVELOPMENT BOARD

FOUR-LAYER TECHNICAL DATASHEET



A custom embedded development platform with USB programming, wireless connectivity, peripheral interfaces and CAN communication.

01 | Product Overview

The ESP32 Development Board is a custom four-layer PCB built around the ESP32-WROOM-32E module. It combines power regulation, USB-to-UART connectivity and common embedded interfaces in a single development platform.

FUNCTION	IMPLEMENTATION
Microcontroller	ESP32-WROOM-32E with integrated Wi-Fi and Bluetooth
USB	USB-C connector with CP2102N USB-to-UART bridge
Power	USB input and AMS1117-3.3 regulator
Display	OLED connector
Communication	UART, SPI, I ² C and CAN
CAN hardware	SN65HVD230 transceiver, NUP2105L protection and selectable 120 Ω termination
Expansion	30-pin, 2.54 mm GPIO header
User controls	Boot and Reset push buttons; user LED

02 | Four-Layer PCB Layout

The following views are taken from the current four-layer KiCad PCB design.

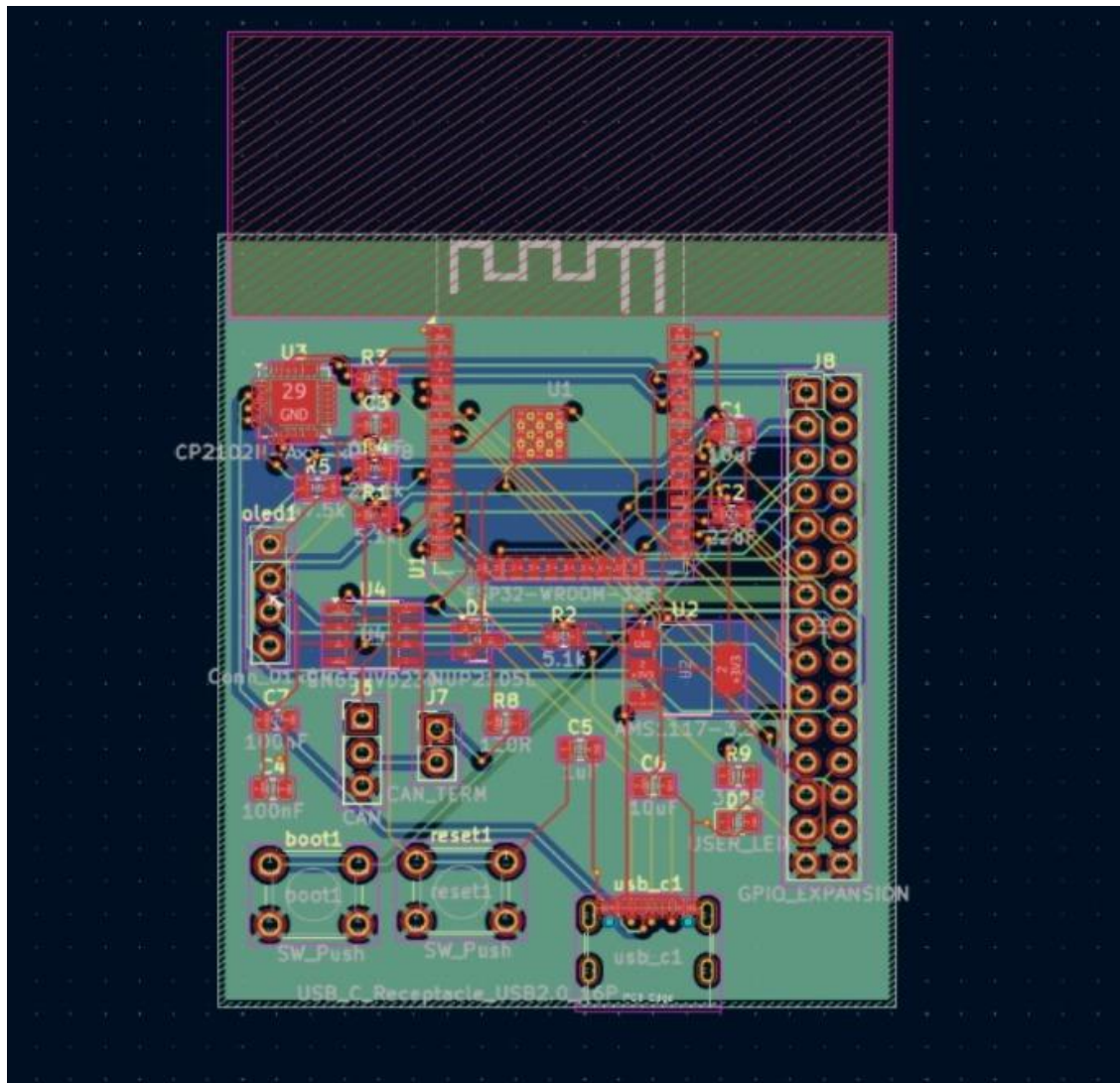
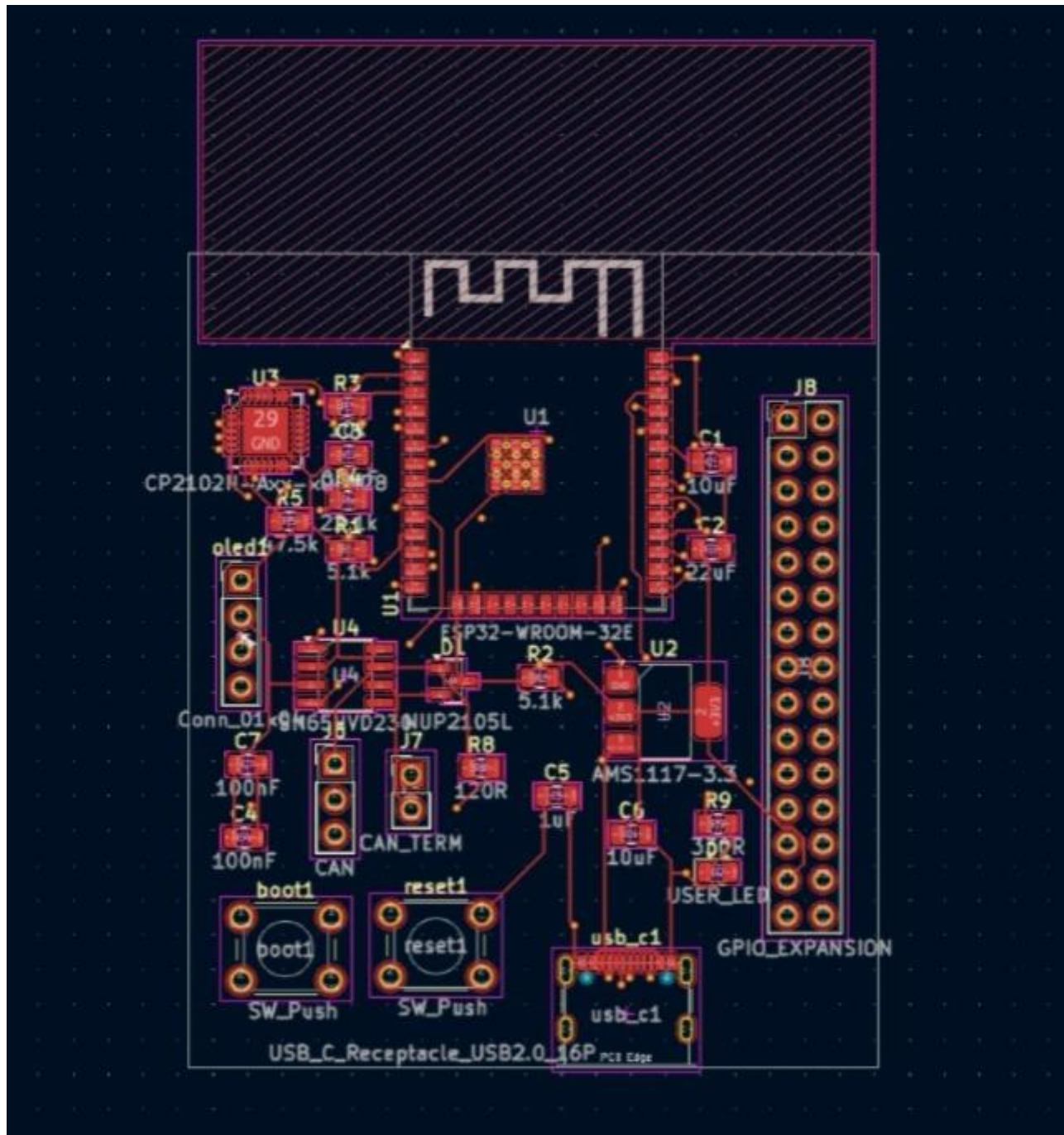


Figure 1. Current PCB layout with F.Cu selected.

03 | F.Cu — Top Copper

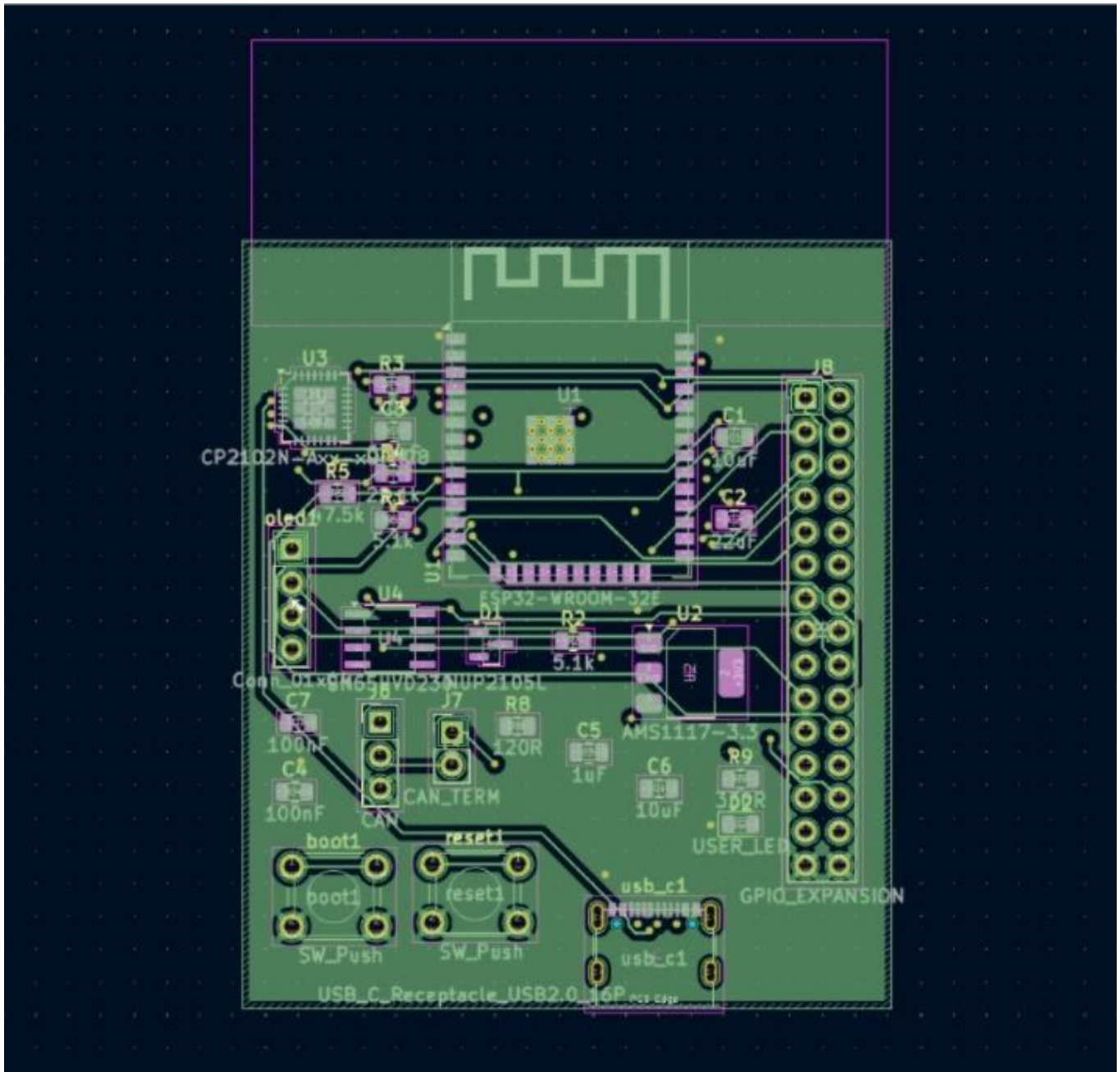
Top copper routing and component-side copper features.



Current four-layer PCB view.

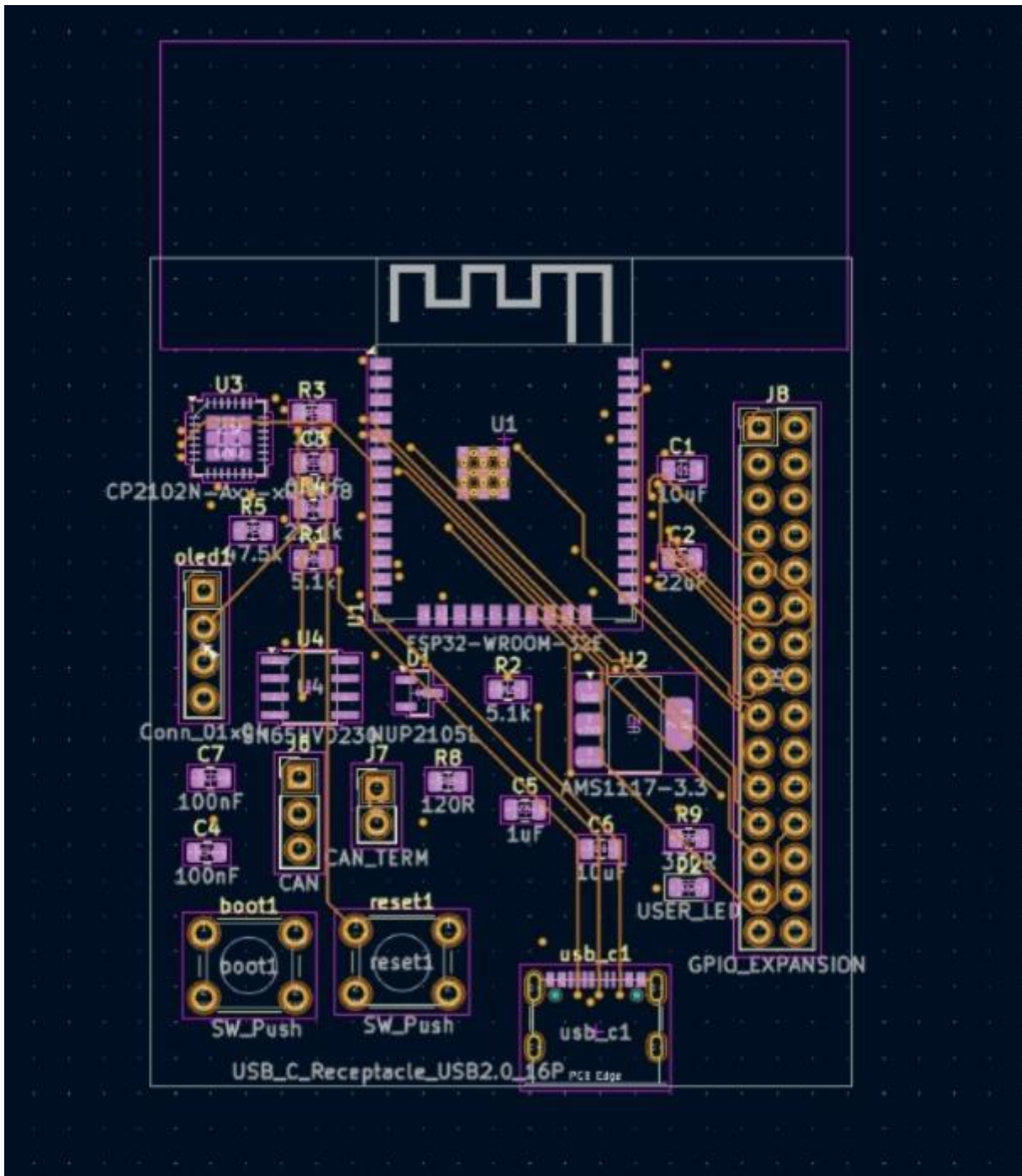
04 | In1.Cu — Inner Ground Layer

Inner copper layer configured as the ground plane.



05 | In2.Cu — Inner Routing Layer

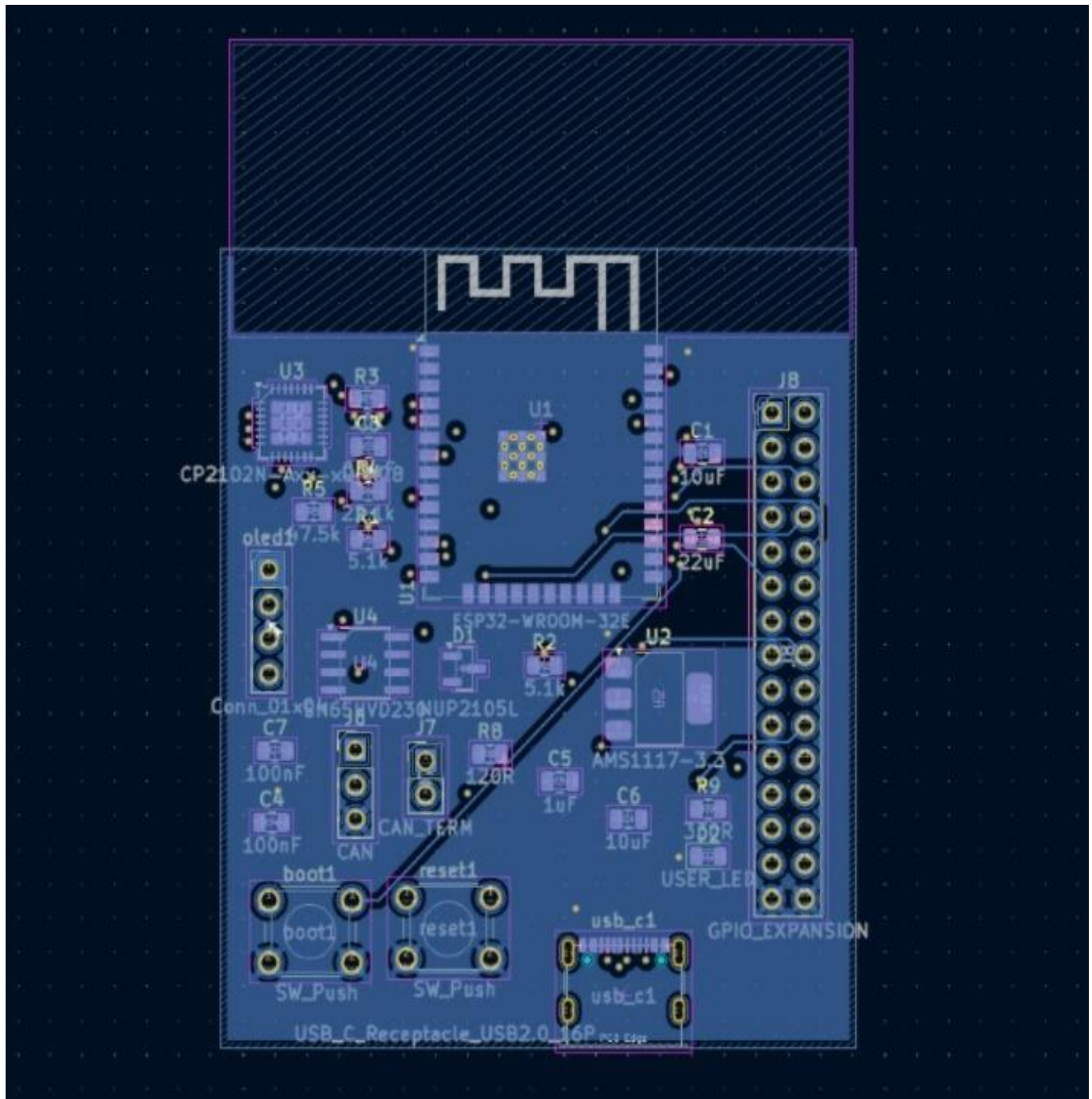
Inner copper layer used for additional signal and power routing.



Current four-layer PCB view.

06 | B.Cu — Bottom Copper

Bottom copper routing and ground copper.



Current four-layer PCB view.

Interface Summary

Schematic

ESP32 Development Board • Four-Layer PCB

08 | GPIO Expansion

J8 is a 30-pin, 2.54 mm-pitch header that exposes the available GPIOs and power connections. GPIO6–GPIO11 are reserved for the module's internal flash interface and are not brought out.

PIN	SIGNAL	PIN	SIGNAL
1	GPIO0	16	GPIO22
2	GPIO1	17	GPIO23
3	GPIO2	18	GPIO25
4	GPIO3	19	GPIO26
5	GPIO4	20	GPIO27
6	GPIO5	21	GPIO32
7	GPIO12	22	GPIO33
8	GPIO13	23	GPIO34
9	GPIO14	24	GPIO35
10	GPIO15	25	GPIO36
11	GPIO16	26	GPIO39
12	GPIO17	27	+3V3
13	GPIO18	28	+5V
14	GPIO19	29	GND
15	GPIO21	30	GND

GPIO34, GPIO35, GPIO36 and GPIO39 are input-only. GPIO0 is used by the boot control circuit. GPIO1 and GPIO3 are shared with the USB-to-UART programming path.

09 | Power and CAN

Power

USB-C supplies the board input. The AMS1117-3.3 regulator generates the 3.3 V rail. Local capacitors provide regulator and device decoupling.

REFERENCE	VALUE / DEVICE	FUNCTION
U2	AMS1117-3.3	3.3 V regulator
C1	1 μ F	Regulator input-side capacitance
C2	22 μ F	Regulator output-side bulk capacitance
C4	100 nF	ESP32/local decoupling
C6	10 μ F	Local bulk decoupling
C7	100 nF	CAN transceiver decoupling

CAN Interface

The CAN interface uses an SN65HVD230 transceiver connected to the ESP32 through GPIO25 and GPIO26. CANH and CANL are available at the external CAN connector. A NUP2105L device provides CAN-line protection.

REFERENCE	DEVICE / VALUE	FUNCTION
U4	SN65HVD230	CAN physical-layer transceiver
NUP2105L	Protection device	CANH / CANL protection
R8	120 Ω	Termination resistor
J7	CAN_TERM	Selectable termination jumper
J6	CAN connector	CANH / CANL / GND

When the termination jumper is installed, the 120 Ω resistor is connected between CANH and CANL. The transceiver RS pin is tied to GND for high-speed mode; Vref is left floating.

10 | PCB Stackup & Design Status

COPPER LAYER	ROLE
F.Cu	Top copper — component-side routing
In1.Cu	Inner copper — GND plane
In2.Cu	Inner copper — additional signal/power routing
B.Cu	Bottom copper — routing and GND copper zone

Configured board thickness: 1.6 mm. Copper thickness: 0.035 mm per copper layer.

Design Verification

ITEM	RECORDED STATUS
Schematic ERC	Completed; no errors or warnings reported
PCB connectivity	0 unconnected items; 0 unrouted connections
Copper zones	Filled
Hardware validation	Fabrication and physical testing have not been performed

Document Information

FIELD	VALUE
Project	ESP32 Development Board
MCU module	ESP32-WROOM-32E
EDA tool	KiCad 9
PCB version	Four-layer
Schematic	Shared with two-layer version
Repository	github.com/Manushree-J/ESP32-Development-Board-PCB

This datasheet documents the PCB design. Electrical performance and operation should be confirmed through fabrication and hardware testing.